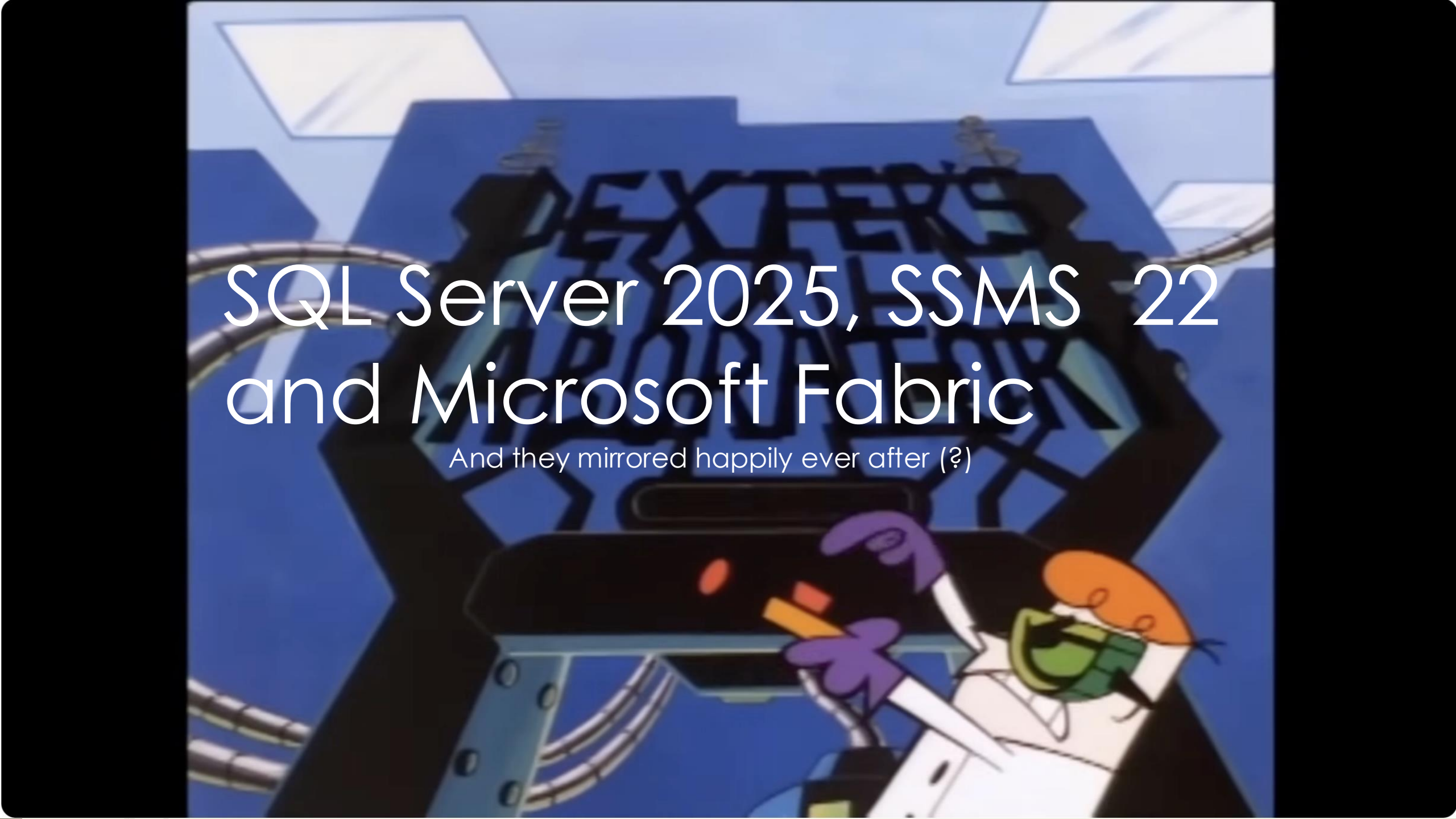


A cartoon illustration of a scientist with orange hair, wearing a white lab coat and green goggles, holding a test tube. In the background, a large blue sign with the words 'DEXTER'S LABORATORY' is visible. The scene is set against a blue sky with white clouds.

SQL Server 2025, SSMS 22 and Microsoft Fabric

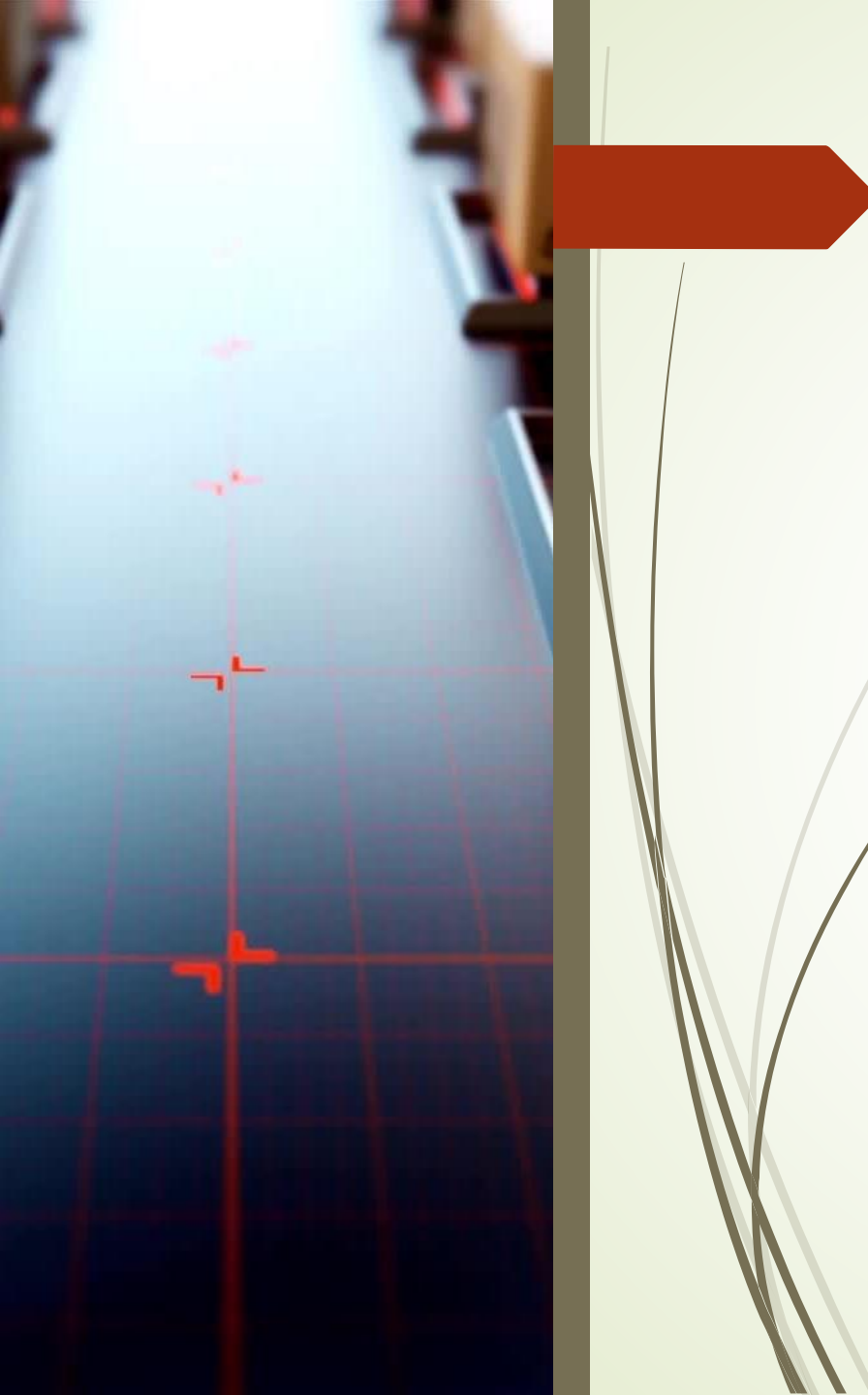
And they mirrored happily ever after (?)



Introduction to mirroring

And other ways of SQL Server data replication





Choices

- High Availability and Disaster Recovery
- Log Shipping
- Managed Instance Link
- Replication
- Change Data Capture

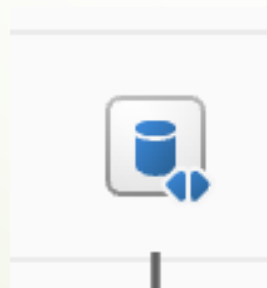
Wait a minute, where's the mirroring gone?

- Deprecated since 2012
- Confusing naming alert!
 - Fabric Mirroring != SQL Mirroring
 - It's called replication in Fabric, but it's not SQL Replication
 - SQL 2016 – 2022: CDC, 2025: change event streaming



New kid on the block

- Fabric mirroring
- Integrate seamlessly into Microsoft Fabric
- Only from read-write databases



Type

Mirrored database

Reitse Eskens

Engineer | Architect

Axians Business Analytics



Reitse.Eskens@axians.com



/in/reitseeskens



<https://sqlreitse.com>

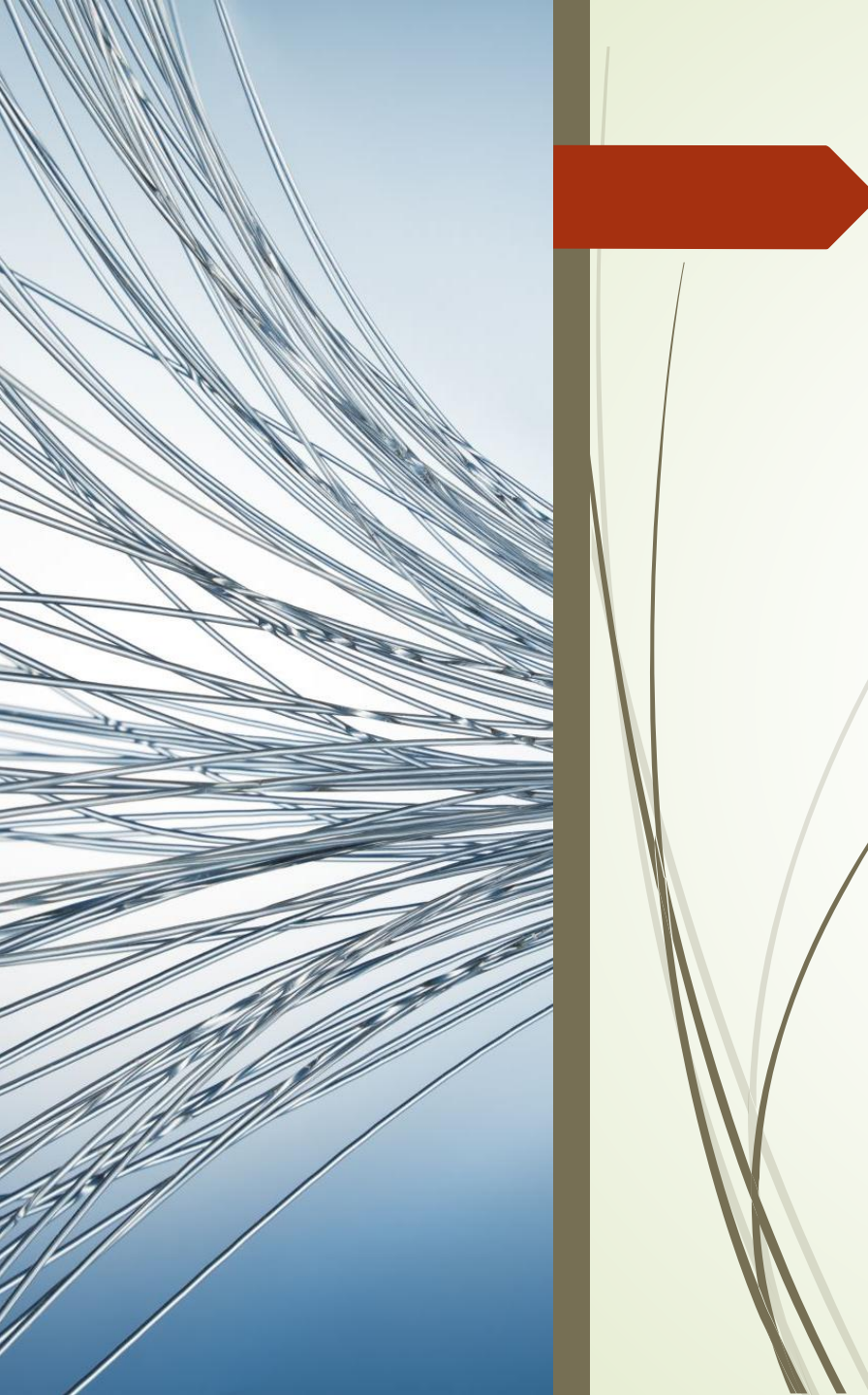


@2meterDBA



A decorative graphic on the left side of the slide. It features a solid red arrow pointing to the right, positioned horizontally. Behind the arrow and extending upwards and to the right are several thin, dark, curved lines that sweep across the frame, creating a sense of movement or flow.

The moving parts



Moving parts

- SQL Server change feed
- Azure Arc
- On-Premises Data Gateway (OPDGW)
- Fabric Mirrored Database

The main setup





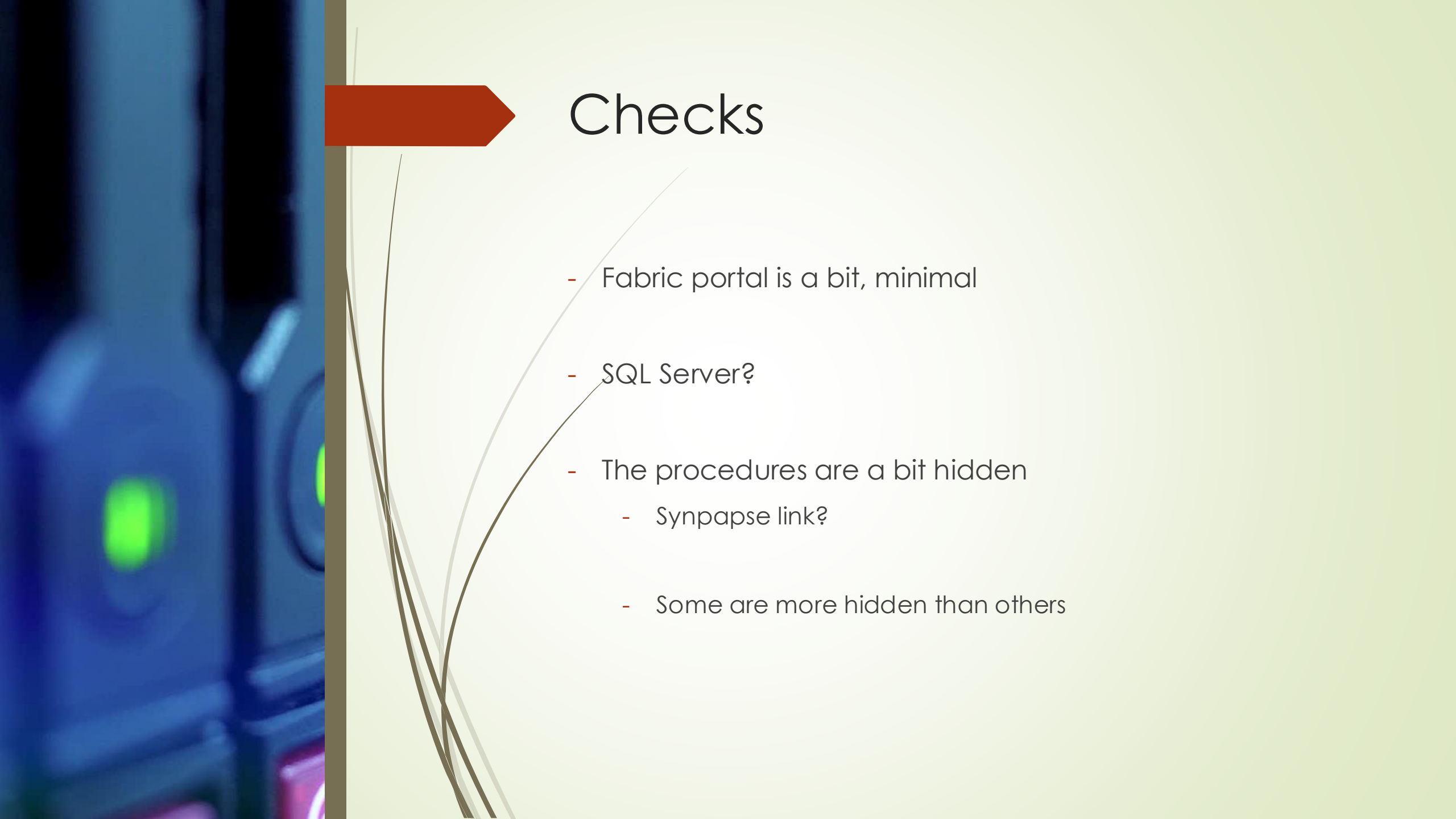
SQL 2016-2022

- Create an account with sufficient privileges (a member of the sysadmin role)
 - Enable CDC
 - Install the On-Premises Data Gateway (OPDG)
 - Create a connection from Fabric to the database
 - Create your mirrored database
-
- Once CDC is set up, enabling Mirroring only requires CONNECT at the server level, and SELECT and CONNECT permissions at the database level to replicate the data.



SQL 2025

- In Azure, start the Azure Arc for SQL process
- Run the PowerShell Script on the server hosting SQL Server
- This creates an Azure Arc resource for the server and the instance(s)
- Enable the System Assigned Managed Identity on the server and the database(!)
- Add the SAMI to the workspace
- Create a connection from Fabric to the database, using the OPDG
- Create your mirrored database resource



Checks

- Fabric portal is a bit, minimal
- SQL Server?
- The procedures are a bit hidden
 - Synpapse link?
 - Some are more hidden than others



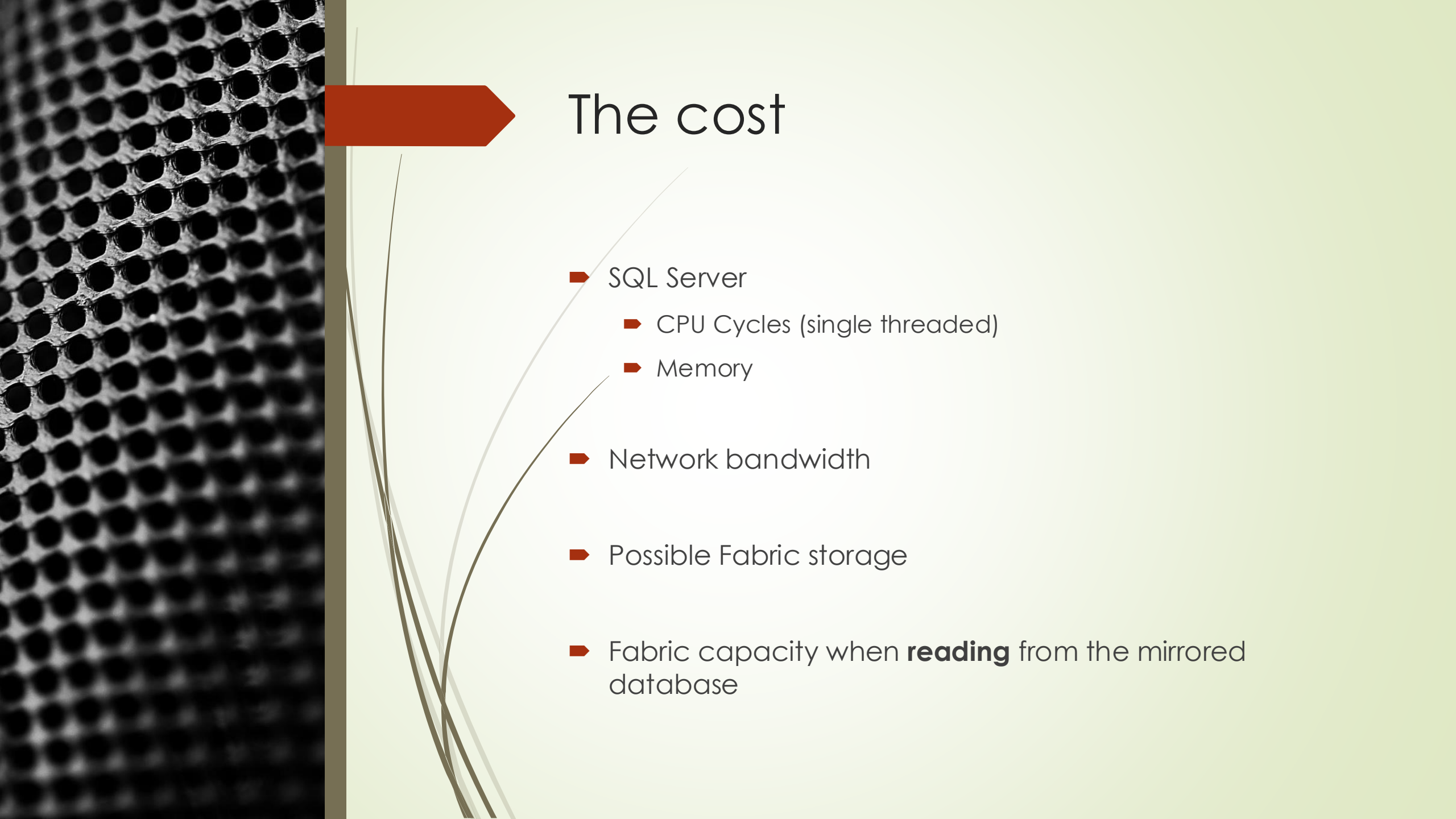
The initial process

- Initiate from Fabric
 - SQL Server sends snapshot to Fabric landing zone
 - Fabric creates parquet with delta
 - Database is mirrored
- 



The update process

- Every 5 seconds, SQL Server checks on changes in the log file
- Changes are pushed to Fabric Landing zone (CSV)
- Fabric reads the CSV and creates the new parquet files
- Files are added to the delta, data is available



The cost

- SQL Server
 - CPU Cycles (single threaded)
 - Memory
- Network bandwidth
- Possible Fabric storage
- Fabric capacity when **reading** from the mirrored database

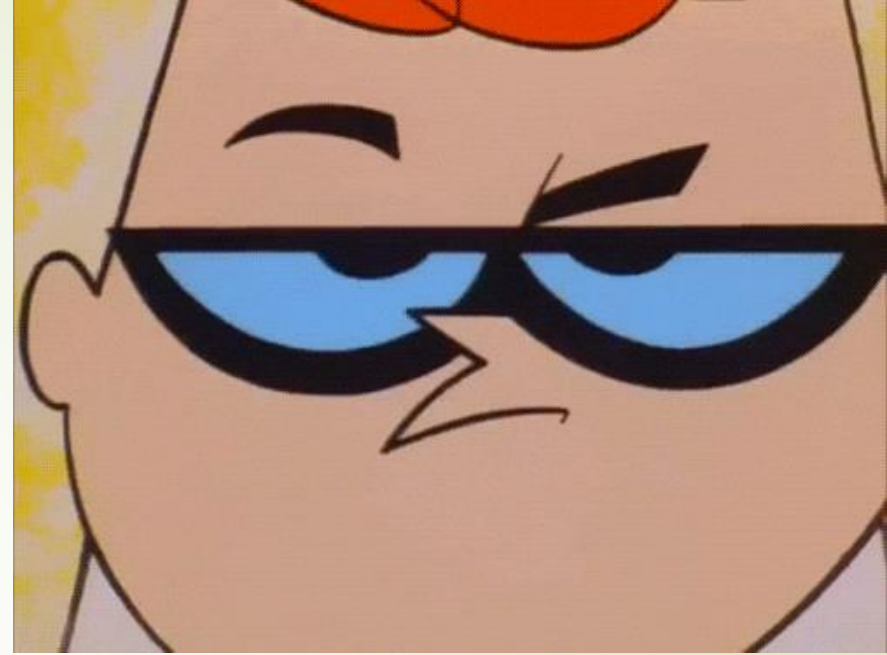


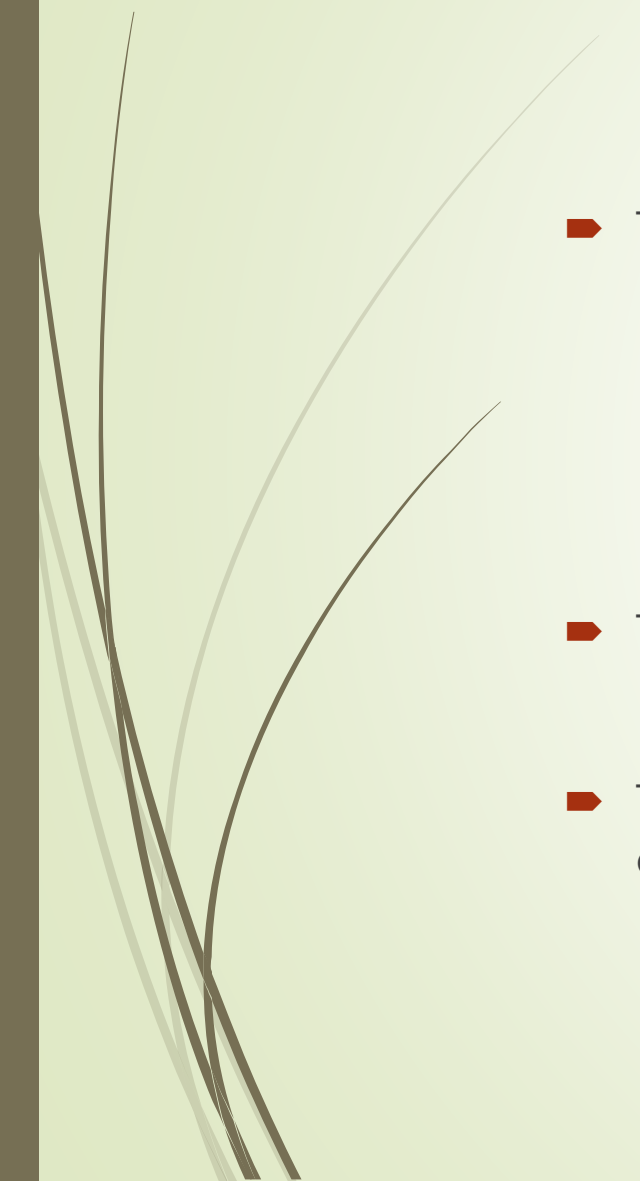
Complexity

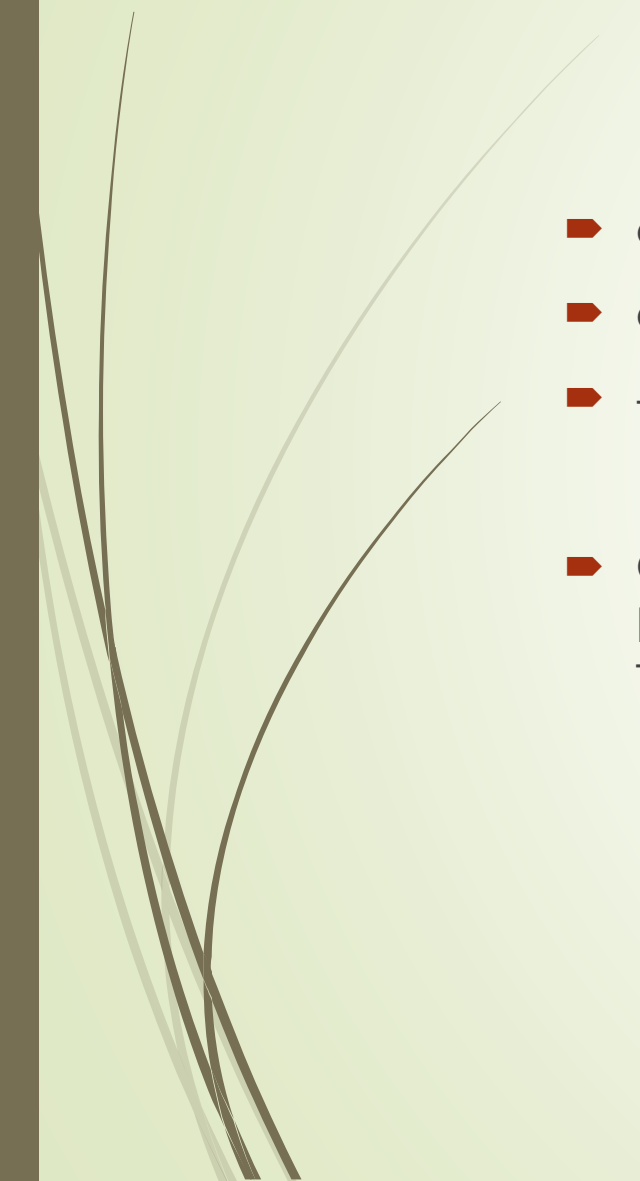
- ▀ Arc
 - ▀ Specific installation requirements
 - ▀ Azure domain admins
- ▀ SQL Server
 - ▀ Highly privileged roles
 - ▀ CISO will have questions (that can be answered)



Security



- 
- 
- To set-up, highly privileged roles necessary
 - Owner on the Azure subscription
 - Administrator on the Fabric Workspace
 - Admin permission on the Server
 - To maintain, someone needs these roles
 - The account used to connect to the database determines the data available

- 
- 
- create login mirrordemo with password = 'P@ssw0rd';
 - create user mirrordemo for login mirrordemo;
 - --Run in the user database

 - GRANT SELECT, ALTER ANY EXTERNAL MIRROR, VIEW DATABASE PERFORMANCE STATE, VIEW DATABASE SECURITY STATE TO mirrordemo;

Demo

Let's do this live (WiFi permitting)



Seeing the moving
parts



SQL Server

- ▶ Extended Events
- ▶ Sys.dm_exec_process
- ▶ Resource governor to limit the impact on your running system
- ▶ DMV's



Microsoft Fabric



OneLake Explorer



Fabric monitor



Limitations

https://learn.microsoft.com/en-us/fabric/mirroring/sql-server-limitations?wt.mc_id=MVP_311370

When it breaks





Restart from Fabric

- It will create a new snapshot
- Increments will be based in that snapshot



Restart from SQL?

- It might not be the best idea
- The system stored procedures are available, but you really have to know what you're doing
- Make sure the full back-ups are running regularly to prevent a full log file
 - Or worse, disk
 - ... and don't put that one on the C:\ drive

Alternatives



Different ways to load data into Fabric

- Copy Job
- Copy data activity
- Dataflow Gen2 (CI/CD enabled)
- DAB + Notebooks
- DBT or other external toolsets

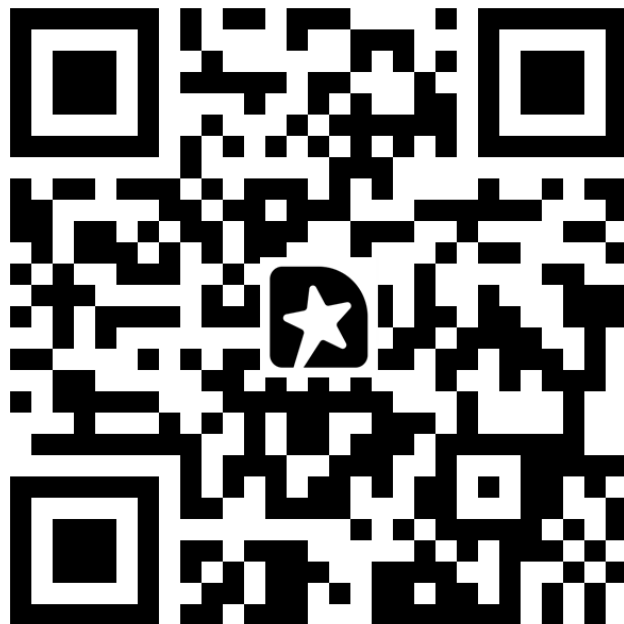
Looking in the mirror





Personal, and professional, opinion

- ▶ Try it out on your dev/test environment
- ▶ Check the load, and decide if it's ok on your system
- ▶ Make an honest comparison with the alternatives
 - ▶ External access
 - ▶ Less predictable, probably heavier loads
 - ▶ Higher Fabric costs



SQL 2025, SSMS 22 and Fabric. And they mirrored happily ever after?



THANK YOU!

Feedback, please 😊